



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-00 Targeted Remediation Cards

Repair one weak competency, then use a changed micro-retry

Artifact	C3_00_TARGETED_REMEDIATION_CARDS_RC1
Status	2026 - LEARNER EDITION
Teaching standard	M01 beginner-first teacher loop; book remains sufficient without video.
Bridge boundary	Course 2B competencies may be assumed; all new engineering concepts still start from first principles.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House rule: understand -> follow once -> see expected result -> break/fix -> do a changed version -> validate -> explain -> save evidence. A running command alone never counts as mastery.



C1 repair - Engineering contract & grain

Weak signal: Input/output grain, rules, failure, rerun and independent validation are explicit.

REPAIR LOOP

1) Restate the exact failed competency. 2) Reopen only the relevant lesson section. 3) Rebuild one tiny worked example. 4) Close material. 5) Complete a changed micro-task. 6) Validate independently. 7) Save as a new repair attempt; never modify the original Gate attempt.

PASS TO RETURN

Changed micro-task is independently correct, validation passes, explain-back is coherent, and no critical failure remains. If the Gate itself needs a full retest, only the Mentor/teacher may assign B or C.



C2 repair - Linux/filesystem safety

Weak signal: Correct shell/path; inspect before destructive action; source/evidence handling safe.

REPAIR LOOP

1) Restate the exact failed competency. 2) Reopen only the relevant lesson section. 3) Rebuild one tiny worked example. 4) Close material. 5) Complete a changed micro-task. 6) Validate independently. 7) Save as a new repair attempt; never modify the original Gate attempt.

PASS TO RETURN

Changed micro-task is independently correct, validation passes, explain-back is coherent, and no critical failure remains. If the Gate itself needs a full retest, only the Mentor/teacher may assign B or C.



C3 repair - Git change hygiene

Weak signal: Reviewable branch/commits; no secret/venv/bytecode; diff understood.

REPAIR LOOP

1) Restate the exact failed competency. 2) Reopen only the relevant lesson section. 3) Rebuild one tiny worked example. 4) Close material. 5) Complete a changed micro-task. 6) Validate independently. 7) Save as a new repair attempt; never modify the original Gate attempt.

PASS TO RETURN

Changed micro-task is independently correct, validation passes, explain-back is coherent, and no critical failure remains. If the Gate itself needs a full retest, only the Mentor/teacher may assign B or C.



C4 repair - Python environment reproducibility

Weak signal: Clean venv/dependency recreation proven.

REPAIR LOOP

1) Restate the exact failed competency. 2) Reopen only the relevant lesson section. 3) Rebuild one tiny worked example. 4) Close material. 5) Complete a changed micro-task. 6) Validate independently. 7) Save as a new repair attempt; never modify the original Gate attempt.

PASS TO RETURN

Changed micro-task is independently correct, validation passes, explain-back is coherent, and no critical failure remains. If the Gate itself needs a full retest, only the Mentor/teacher may assign B or C.



C5 repair - Structure/config/secret boundary

Weak signal: Core/CLI boundary + documented config precedence + no secret leakage.

REPAIR LOOP

1) Restate the exact failed competency. 2) Reopen only the relevant lesson section. 3) Rebuild one tiny worked example. 4) Close material. 5) Complete a changed micro-task. 6) Validate independently. 7) Save as a new repair attempt; never modify the original Gate attempt.

PASS TO RETURN

Changed micro-task is independently correct, validation passes, explain-back is coherent, and no critical failure remains. If the Gate itself needs a full retest, only the Mentor/teacher may assign B or C.



C6 repair - Logging/failure semantics

Weak signal: Actionable logs; known failure non-zero; unexpected error not swallowed.

REPAIR LOOP

1) Restate the exact failed competency. 2) Reopen only the relevant lesson section. 3) Rebuild one tiny worked example. 4) Close material. 5) Complete a changed micro-task. 6) Validate independently. 7) Save as a new repair attempt; never modify the original Gate attempt.

PASS TO RETURN

Changed micro-task is independently correct, validation passes, explain-back is coherent, and no critical failure remains. If the Gate itself needs a full retest, only the Mentor/teacher may assign B or C.



C7 repair - Testing

Weak signal: Tests protect contract incl. failure/edge case; deliberate regression is catchable.

REPAIR LOOP

1) Restate the exact failed competency. 2) Reopen only the relevant lesson section. 3) Rebuild one tiny worked example. 4) Close material. 5) Complete a changed micro-task. 6) Validate independently. 7) Save as a new repair attempt; never modify the original Gate attempt.

PASS TO RETURN

Changed micro-task is independently correct, validation passes, explain-back is coherent, and no critical failure remains. If the Gate itself needs a full retest, only the Mentor/teacher may assign B or C.



C8 repair - CLI contract

Weak signal: Useful help, stable invocation/output and machine-readable failure behavior.

REPAIR LOOP

1) Restate the exact failed competency. 2) Reopen only the relevant lesson section. 3) Rebuild one tiny worked example. 4) Close material. 5) Complete a changed micro-task. 6) Validate independently. 7) Save as a new repair attempt; never modify the original Gate attempt.

PASS TO RETURN

Changed micro-task is independently correct, validation passes, explain-back is coherent, and no critical failure remains. If the Gate itself needs a full retest, only the Mentor/teacher may assign B or C.



C9 repair - Container reasoning

Weak signal: Image/container/mount/runtime reasoning correct; real Docker evidence or honest approved fallback.

REPAIR LOOP

1) Restate the exact failed competency. 2) Reopen only the relevant lesson section. 3) Rebuild one tiny worked example. 4) Close material. 5) Complete a changed micro-task. 6) Validate independently. 7) Save as a new repair attempt; never modify the original Gate attempt.

PASS TO RETURN

Changed micro-task is independently correct, validation passes, explain-back is coherent, and no critical failure remains. If the Gate itself needs a full retest, only the Mentor/teacher may assign B or C.



C10 repair - Idempotence/reproducibility/runbook

Weak signal: Safe rerun demonstrated; operator can reproduce/validate/recover from runbook.

REPAIR LOOP

1) Restate the exact failed competency. 2) Reopen only the relevant lesson section. 3) Rebuild one tiny worked example. 4) Close material. 5) Complete a changed micro-task. 6) Validate independently. 7) Save as a new repair attempt; never modify the original Gate attempt.

PASS TO RETURN

Changed micro-task is independently correct, validation passes, explain-back is coherent, and no critical failure remains. If the Gate itself needs a full retest, only the Mentor/teacher may assign B or C.